Four Fundamental Subspaces

Goal: Solve & Analyze solution of

$$(*) \qquad Ax = b \quad \text{for} \quad x \in V$$

where $A: V \to W$, $b \in W$ are given.

The **nullspace** and the **range** of A

$$N(A) = \{u \in V \mid Au = 0\} \subseteq V$$

$$A(\alpha u_1 + u_2) = \alpha A u_1 + A u_2 = 0$$

$$R(A) = \{Au \mid u \in V\} \subseteq W$$

$$\alpha b_1 + b_2 = \alpha A u_1 + A u_2 = A(\alpha u_1 + u_2)$$

Criterion for Well-Posed (*):

1) Solution Exists
2) Solution is Unique
3) Solution depends continuously on A, b

1) $b \in R(A) \subseteq W$
2) $N(A) = \{0\} \subset V$
3) "Bounded Invertibility"

Example: $\frac{d}{dx} : C^\infty_{per}(0,1) \to C^\infty_{per}(0,1)$

$$\frac{d}{dx} u = f \quad \to \quad u(x) = c + \int_0^x f(s)\,ds$$

Only periodic if $u(1) - u(0) = \int_0^1 f(s)\,ds$.

$$\Rightarrow R\left(\frac{d}{dx}\right) \subseteq \left\{ f \in C^\infty_{per}(0,1) \;\middle|\; \int_0^1 f(s)\,ds = 0 \right\}$$

equivalently

$$\Rightarrow R\left(\frac{d}{dx}\right) \perp \left\{ f \in C^\infty_{per}(0,1) \;\middle|\; f = \text{const.} \right\}$$

Differentiation in the Fourier Basis

Consider the Fourier basis for $L^2(0,1) \supset C^\infty_{per}(0,1)$

$$e_k(x) = e^{2\pi i k x}, \qquad k = 0, \pm 1, \pm 2, \pm 3, \ldots$$

For $f \in C^\infty_{per}(0,1)$, we can write

$$f(x) = \sum_{k=-\infty}^{+\infty} \hat{f}_k e^{2\pi i k x}, \qquad \hat{f}_k = \int_0^1 f(x) e^{-2\pi i k x}\,dx$$

Since $|\hat{f}_k|$ decay faster than $O(|k|^{-p})$ for any $p = 0, 1, 2, \ldots$

$$\frac{d}{dx} f(x) = \sum_{k=-\infty}^{+\infty} (2\pi i k) \hat{f}_k e^{2\pi i k x}$$

$$\vdots$$

$$\frac{d^p}{dx^p} f(x) = \sum_{k=-\infty}^{+\infty} (2\pi i k)^p \hat{f}_k e^{2\pi i k x}$$

As a linear transformation on $C^\infty_{per}(0,1)$,

$$\sum_{k=-\infty}^{+\infty} \hat{f}_k e^{2\pi i k x} \xrightarrow{\frac{d^p}{dx^p}} \sum_{k=-\infty}^{+\infty} (2\pi i k)^p \hat{f}_k e^{2\pi i k x}$$

$$\begin{bmatrix} \ddots & & \\ & (-2\pi i)^p & \\ & & (2\pi i)^p \\ & & & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \hat{f}_{-1} \\ \hat{f}_0 \\ \hat{f}_1 \\ \vdots \end{bmatrix} = (2\pi i)^p \begin{bmatrix} \vdots \\ (-1)^p \hat{f}_{-1} \\ 0 \cdot \hat{f}_0 \\ (+1)^p \hat{f}_{+1} \\ \vdots \end{bmatrix}$$

D $\qquad$ $\underline{\hat{f}}$ $\qquad$ $\underline{\left(\widehat{\frac{d^p f}{dx^p}}\right)}$

$$D_{jk} = \begin{cases} (2\pi i k)^p & j = k \\ 0 & j \neq k \end{cases}$$

$$\Rightarrow R\left(\frac{d}{dx}\right) = \left\{ f \in C^\infty_{per}(0,1) \;\middle|\; \int_0^1 f(s)\,ds = 0 \right\} \quad \text{dim} = \infty$$

$$\Rightarrow N\left(\frac{d}{dx}\right) = \left\{ f \in C^\infty_{per}(0,1) \;\middle|\; f = \text{const.} \right\} \quad \text{dim} = 1$$

The Adjoint of a Differential Operator

For a linear operator $A: V \to V$, where V is equipped w/ inner product $\langle \cdot, \cdot \rangle$, the adjoint of A is characterized by

$$\langle Au, v \rangle = \langle u, A^* v \rangle \qquad u, v \in V.$$

Example: $V = \mathbb{C}^n$, $A \in \mathbb{C}^{n \times n}$ (matrix)

$$\langle Au, v \rangle = v^*(Au) = (v^*A)u \overset{?}{=} \langle u, A^* v \rangle$$

$$[\ ][\ ][\ ] \qquad (A^*v)^* u$$

Example: $V = C^\infty_{per}(0,1)$, $A = \frac{d}{dx}$

$$\langle Au, v \rangle = \int_0^1 \left(\frac{du}{dx}\right) \bar{v}\, dx = u\bar{v}\Big|_0^1 - \int_0^1 u \frac{d\bar{v}}{dx}\, dx$$

($u\bar{v}\big|_0^1 = 0$)

$$A^* = -\frac{d}{dx} \qquad = \langle u, A^* v \rangle$$

The Four Fundamental Subspaces

The 2 fundamental subspaces of A are closely related to the 2 F.S. of A^*.

Suppose $f = Au \in R(A)$ and $v \in N(A^*)$.

$$\langle f, v \rangle = \langle Au, v \rangle = \langle u, A^* v \rangle = \langle u, 0 \rangle = 0$$

$$f \in R(A) \Rightarrow f \perp N(A^*)$$
$$f \in R(A^*) \Rightarrow f \perp N(A)$$

When $\dim(V) < \infty$, this is

"The Fundamental Theorem of Linear Algebra"
- Gil Strang

Constant Coeff. Diff. Ops

$$A = a_0 + a_1 \frac{d}{dx} + \cdots + a_n \frac{d^n}{dx^n} \qquad a_n \neq 0.$$

$$Ae_k = \underbrace{[a_0 + (2\pi i k)^1 a_1 + \cdots + (2\pi i k)^n a_n]}_{p(2\pi i k)} e_k$$

characteristic polynomial

$$f(x) = \sum_{k=-\infty}^{+\infty} \hat{f}_k e^{2\pi i k x} \rightarrow [Af](x) = \sum_{k=-\infty}^{+\infty} p(2\pi i k) \hat{f}_k e^{2\pi i k x}$$